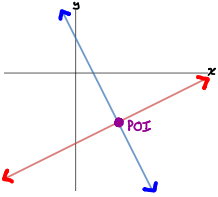
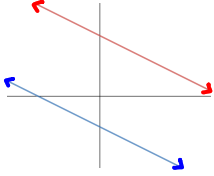
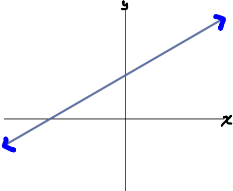


Remember that _____ a linear system means to find the values of the variables that satisfy ALL of the equations in the system. Graphically speaking, this means you will find the ordered pair (x, y) where the lines intersect.

There are 3 main methods for solving a linear system:

- 1) Graphing
- 2) Substitution
- 3) Elimination

A linear system could have 1, 0, or infinitely many solutions:

Graph	Slopes of Lines	Intercepts	Number of Solutions	What happens algebraically
Intersecting 				
Parallel & Distinct 				
Parallel & Coincident 				

Steps for Solving by Substitution:

- 1) Rearrange either equation to isolate a variable (x or y)
- 2) Substitute what the isolated variable is equal to into the OTHER equation
- 3) Solve the new equation for the variable
- 4) Plug your answer back in to EITHER original equation to solve for the OTHER variable.
- 5) Check your answer in BOTH equations

Example 1: Solve the following systems using the method of substitution

a) $\ell_1: x + 4y = 6$
 $\ell_2: 2x - 3y = 1$

b) $\ell_1: 5x - 3y - 2 = 0$
 $\ell_2: 7x + y = 0$

c) $\ell_1: 2x + 2y = 7$
 $\ell_2: x + y = 6$

d) $\ell_1: 3x + 4y = 2$
 $\ell_2: 9x + 12y = 6$